MID SEMESTER PRACTICAL EXAMINATION

You are required to act as a Business Intelligence Analyst solving a real-world organizational problem using Power BI.

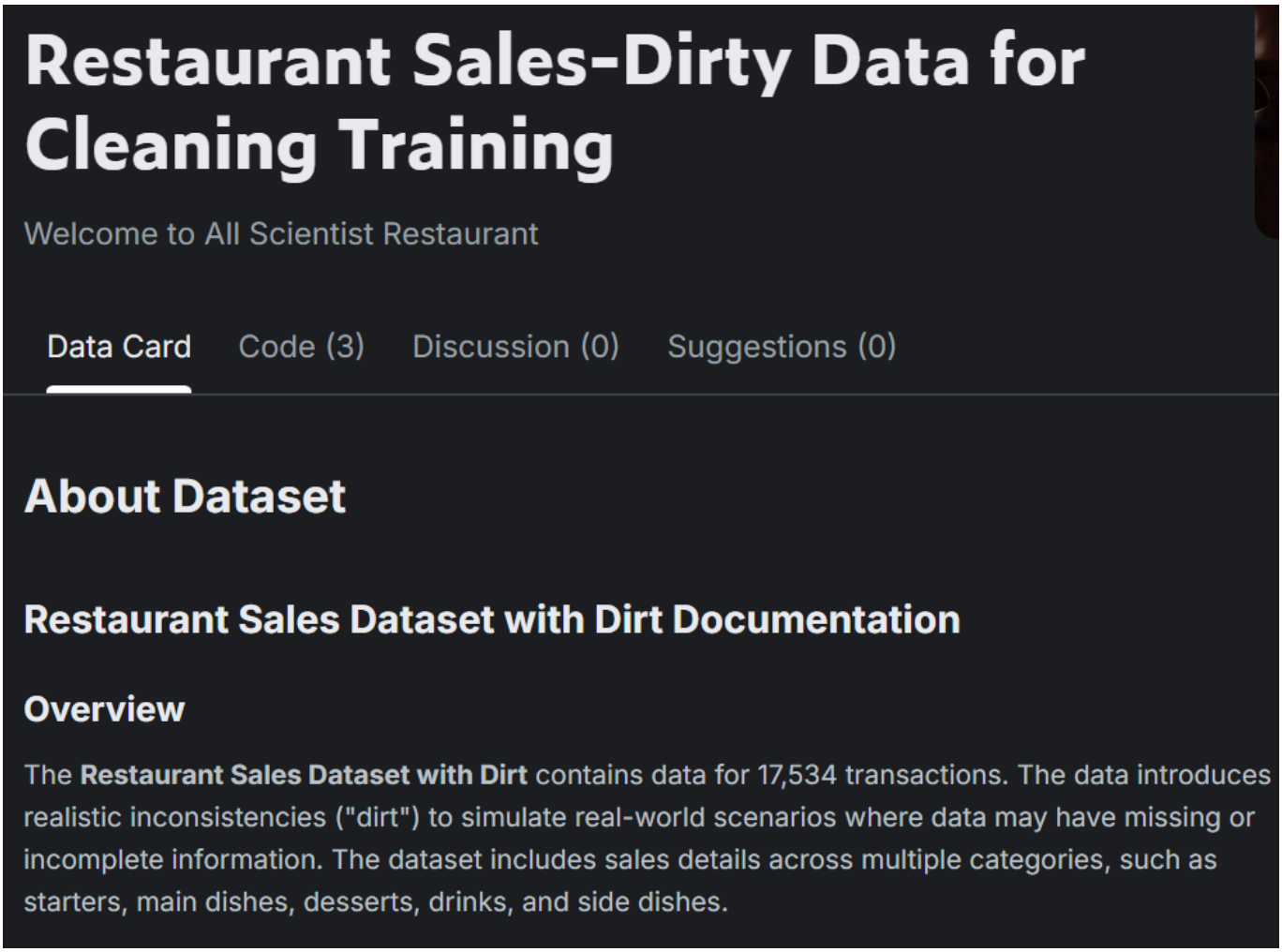
You must complete a full end-to-end BI workflow: Dataset Selection → Data Preparation → Data Modelling → Dashboard Development → Publishing

Dataset Selection

My dataset is from Kaggle, and it is called `"Restaurant Sales"`. The Restaurant Sales Dataset with Dirt contains data for `17,534` transactions.

You can find the dataset here.

Details about the dataset can be found in the following image:



Columns Description

Column Name	Description	Example Values
Order ID	A unique identifier for each order.	ORD_123456

Column Name	Description	Example Values
Customer ID	A unique identifier for each customer.	CUST_001
Category	The category of the purchased item.	Main Dishes, Drinks
Item	The name of the purchased item. May contain missing values due to data dirt.	Grilled Chicken, None
Price	The static price of the item. May contain missing values.	15.0, None
Quantity	The quantity of the purchased item. May contain missing values.	1, None
Order Total	The total price for the order (Price × Quantity). May contain missing values.	45.0, None
Order Date	The date when the order was placed. Always present.	2022-01-15
Payment Method	The payment method used for the transaction. May contain missing values.	Cash, None

A sample of the dataset is shown below:

Order ID A unique identifier for each order.	Customer ID A unique identifier for each customer.	Category The category of the purchased item.	Item The name of the purchased item.	Price The static price of the item.	Quantity The quantity of the purchased item.	Order Total The total price for the order (`Price * Quantity`)	Order Date The date when the order was placed.	Payment Method The payment method used for the transaction.
17534 unique values	CUST_066 1% CUST_100 1% Other (17127) 98%	Main Dishes 20% Starters 20% Other (10456) 60%	[null] 10% Pasta Alfredo 6% Other (14778) 84%	1 20	1 5	1 100	2022-01-01 2023-12-31	Credit Card 31% Cash 31% Other (6531) 37%
ORD_705844	CUST_092	Side Dishes	Side Salad	3.0	1.0	3.0	2023-12-21	Credit Card
ORD_338528	CUST_021	Side Dishes	Mashed Potatoes	4.0	3.0	12.0	2023-05-19	Digital Wallet
ORD_443849	CUST_029	Main Dishes	Grilled Chicken	15.0	4.0	60.0	2023-09-27	Credit Card
ORD_630508	CUST_075	Drinks			2.0	5.0	2022-08-09	Credit Card
ORD_648269	CUST_031	Main Dishes	Pasta Alfredo	12.0	4.0	48.0	2022-05-15	Cash
ORD_381680	CUST_031	Main Dishes	Salmon	18.0	5.0	90.0	2022-07-20	Digital Wallet
ORD_270994	CUST_071	Side Dishes	Garlic Bread	4.0	5.0	20.0	2022-08-19	Credit Card
ORD_146656	CUST_077	Main Dishes		15.0	3.0	45.0	2023-02-15	Cash
ORD_428611	CUST_083	Desserts		6.0	2.0	12.0	2023-12-16	Cash
ORD_743636	CUST_085	Main Dishes	Vegetarian Platter	14.0	5.0	70.0	2022-08-07	

To obtain multiple tables, I split the dataset into two tables: `Order Table` and `Customer Table`.

```
import pandas as pd
import numpy as np

# Load the dataset
data = pd.read_csv('restaurant_sales_data.csv')

# Splitting the data
table1 = data[
    ['Order ID', 'Order Total', 'Order Date', 'Payment Method']
]

table2 = data[
    ['Customer ID', 'Category', 'Item', 'Price', 'Quantity', 'Order ID']
]
```

```
# Save the tables to new CSV files
table1.to_csv('Order Table.csv', index=False)
table2.to_csv('Customer Table.csv', index=False)
```

The `Order Table` contains the following columns: 'Order ID', 'Order Total', 'Order Date', 'Payment Method'

Order ID	Order Total	Order Date	Payment Method
Missing: 0 (0%)	Missing: 430 (2%)	Missing: 0 (0%)	Missing: 1082 (6%)
Distinct: 17534 (100%)	Distinct: 41 (<1%)	Distinct: 730 (4%)	Distinct: 3 (<1%)
17534 Distinct values	Min 1.0 Max 100.0	730 Distinct values	Credit Card 31% Cash 31% Digital Wallet 31%

	Order ID	Order Total	Order Date	Payment Method
0	ORD_705844	3.0	2023-12-21	Credit Card
1	ORD_338528	12.0	2023-05-19	Digital Wallet
2	ORD_443849	60.0	2023-09-27	Credit Card
3	ORD_630508	5.0	2022-08-09	Credit Card
4	ORD_648269	48.0	2022-05-15	Cash
5	ORD_381680	90.0	2022-07-20	Digital Wallet
6	ORD_270994	20.0	2022-08-19	Credit Card
7	ORD_146656	45.0	2023-02-15	Cash
8	ORD_428611	12.0	2023-12-16	Cash
9	ORD_743636	70.0	2022-08-07	*Missing value*
10	ORD_117003	12.0	2023-12-09	Credit Card

The `Customer Table` contains the following columns: 'Customer ID', 'Category', 'Item', 'Price', 'Quantity', 'Order ID'

Customer ID	Category	Item	Price	Quantity	Order ID
Missing: 0 (0%)	Missing: 0 (0%)	Missing: 1758 (10%)	Missing: 876 (5%)	Missing: 430 (2%)	Missing: 0 (0
Distinct: 100 (<1%)	Distinct: 5 (<1%)	Distinct: 26 (<1%)	Distinct: 14 (<1%)	Distinct: 5 (<1%)	Distinct: 17534 (100
100 Distinct values	Main Dishes 20% Starters 20% Desserts 20% Other 40%	26 Distinct values	Min 1.0 Max 20.0	Min 1.0 Max 5.0	17534 Distinct values

	Customer ID	Category	Item	Price	Quantity	Order ID
0	CUST_092	Side Dishes	Side Salad	3.0	1.0	ORD_705844
1	CUST_021	Side Dishes	Mashed Potatoes	4.0	3.0	ORD_338528
2	CUST_029	Main Dishes	Grilled Chicken	15.0	4.0	ORD_443849
3	CUST_075	Drinks	*Missing value*	*Missing value*	2.0	ORD_630508
4	CUST_031	Main Dishes	Pasta Alfredo	12.0	4.0	ORD_648269
5	CUST_031	Main Dishes	Salmon	18.0	5.0	ORD_381680
6	CUST_071	Side Dishes	Garlic Bread	4.0	5.0	ORD_270994
7	CUST_077	Main Dishes	*Missing value*	15.0	3.0	ORD_146656
8	CUST_083	Desserts	*Missing value*	6.0	2.0	ORD_428611
9	CUST_085	Main Dishes	Vegetarian Platter	14.0	5.0	ORD_743636
10	CUST_021	Desserts	Brownie	6.0	2.0	ORD_117003
11	CUST_025	Main Dishes	Salmon	18.0	5.0	ORD_726419
12	CUST_047	Main Dishes	Salmon	18.0	1.0	ORD_754176

Part A: Power Query Editor(Data Preparation)

Loading the data in Power BI

- Loading process of the Customer Table

Customer Table.csv

File Origin: 1252: Western European (Windows)

Delimiter: Comma

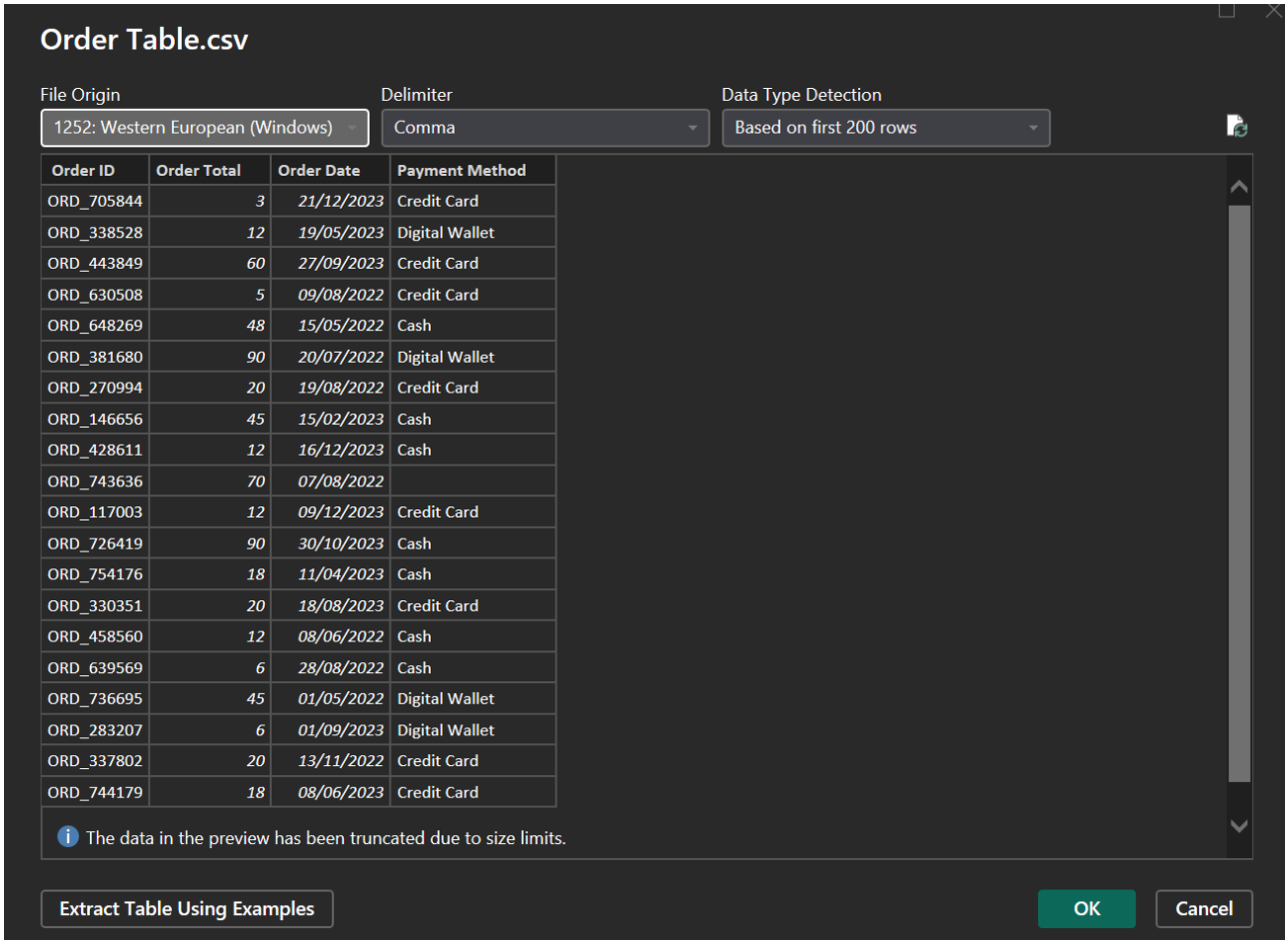
Data Type Detection: Based on first 200 rows

Customer ID	Category	Item	Price	Quantity	Order ID
CUST_092	Side Dishes	Side Salad	3	1	ORD_705844
CUST_021	Side Dishes	Mashed Potatoes	4	3	ORD_338528
CUST_029	Main Dishes	Grilled Chicken	15	4	ORD_443849
CUST_075	Drinks		null	2	ORD_630508
CUST_031	Main Dishes	Pasta Alfredo	12	4	ORD_648269
CUST_031	Main Dishes	Salmon	18	5	ORD_381680
CUST_071	Side Dishes	Garlic Bread	4	5	ORD_270994
CUST_077	Main Dishes		15	3	ORD_146656
CUST_083	Desserts		6	2	ORD_428611
CUST_085	Main Dishes	Vegetarian Platter	14	5	ORD_743636
CUST_021	Desserts	Brownie	6	2	ORD_117003
CUST_025	Main Dishes	Salmon	18	5	ORD_726419
CUST_047	Main Dishes	Salmon	18	1	ORD_754176
CUST_057	Starters	Sweet Potato Fries	5	4	ORD_330351
CUST_017	Side Dishes	Mashed Potatoes	4	3	ORD_458560
CUST_024	Drinks	Lemonade	3	2	ORD_639569
CUST_067	Main Dishes	Grilled Chicken	15	3	ORD_736695
CUST_090	Side Dishes	Side Salad	3	2	ORD_283207
CUST_034	Desserts		4	5	ORD_337802
CUST_009	Desserts	Brownie	6	3	ORD_744179

The data in the preview has been truncated due to size limits.

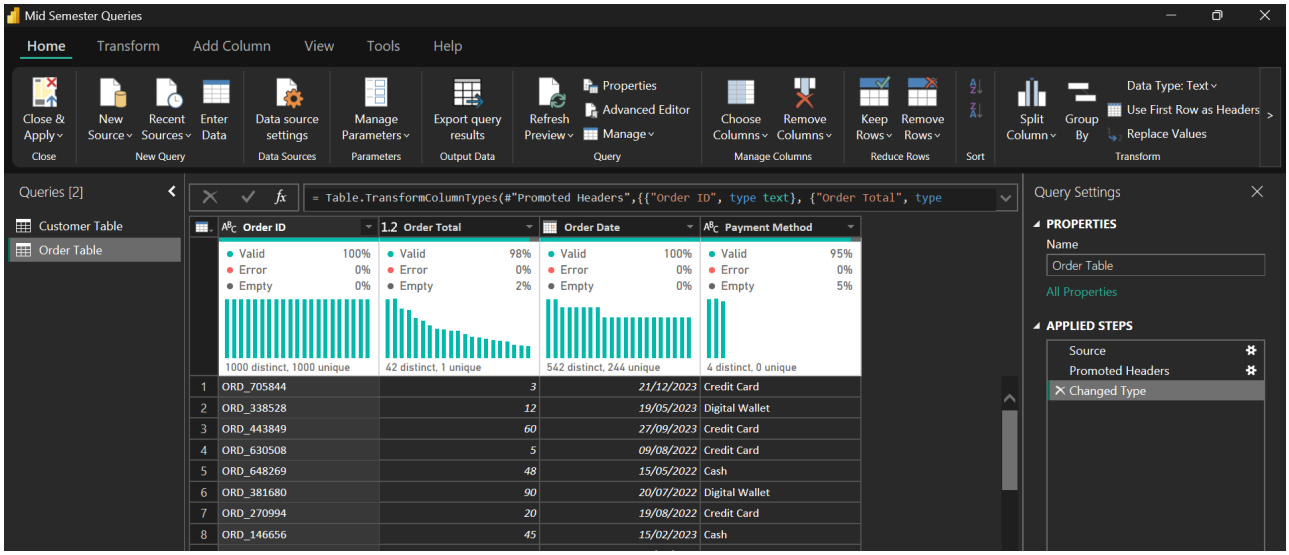
Extract Table Using Examples | Load | Transform Data | Cancel

- Loading process of the Order Table



Order ID	Order Total	Order Date	Payment Method
ORD_705844	3	21/12/2023	Credit Card
ORD_338528	12	19/05/2023	Digital Wallet
ORD_443849	60	27/09/2023	Credit Card
ORD_630508	5	09/08/2022	Credit Card
ORD_648269	48	15/05/2022	Cash
ORD_381680	90	20/07/2022	Digital Wallet
ORD_270994	20	19/08/2022	Credit Card
ORD_146656	45	15/02/2023	Cash
ORD_428611	12	16/12/2023	Cash
ORD_743636	70	07/08/2022	
ORD_117003	12	09/12/2023	Credit Card
ORD_726419	90	30/10/2023	Cash
ORD_754176	18	11/04/2023	Cash
ORD_330351	20	18/08/2023	Credit Card
ORD_458560	12	08/06/2022	Cash
ORD_639569	6	28/08/2022	Cash
ORD_736695	45	01/05/2022	Digital Wallet
ORD_283207	6	01/09/2023	Digital Wallet
ORD_337802	20	13/11/2022	Credit Card
ORD_744179	18	08/06/2023	Credit Card

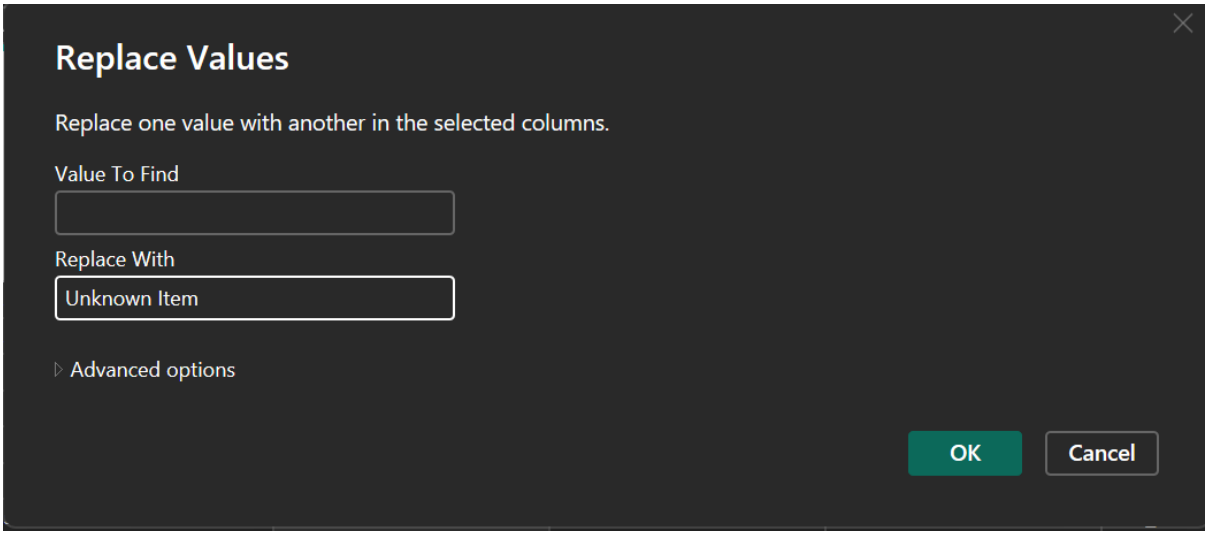
- Power Query Editor after loading the tables



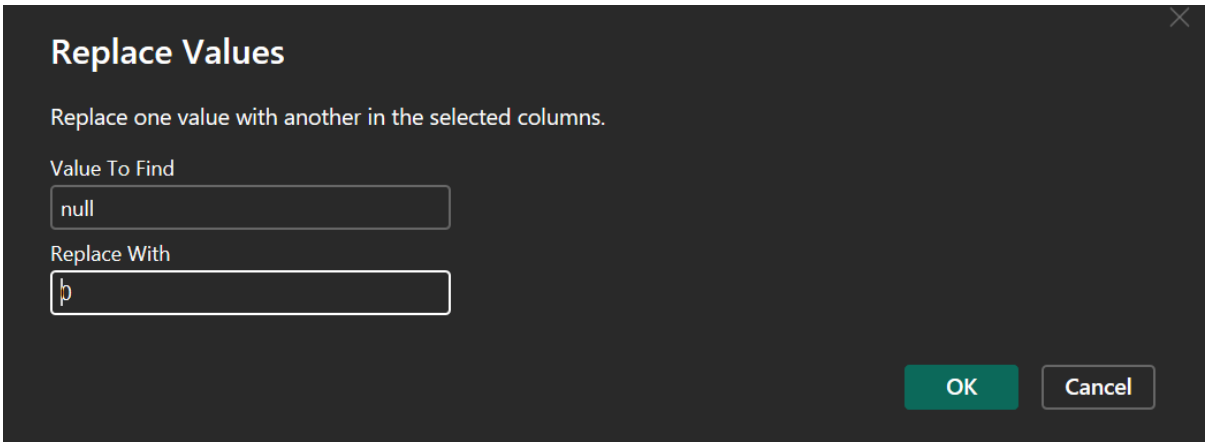
Data Cleaning and Transformation

- Removing duplicates in the Customer Table
- Handling missing or null values

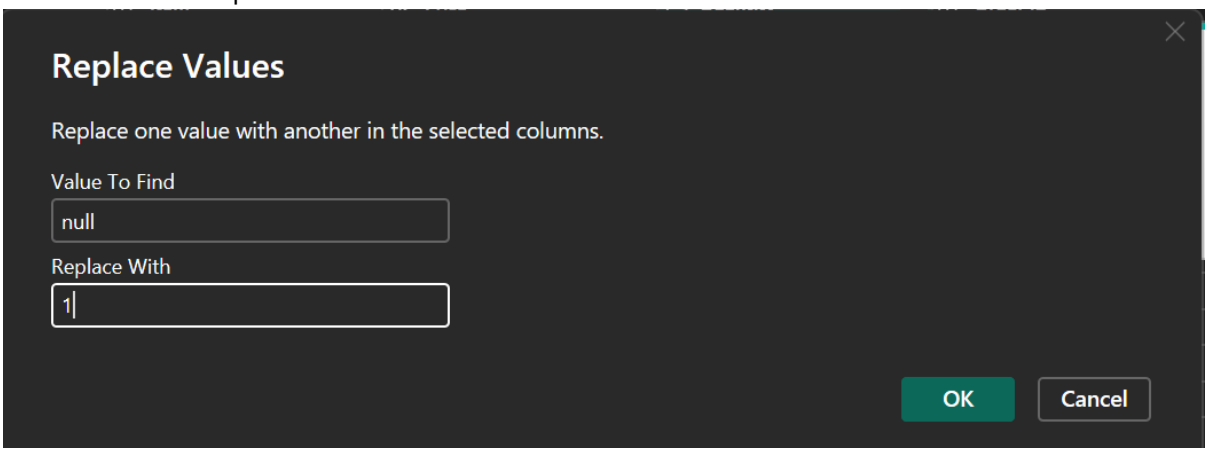
- Filled missing Items and Payment Method with "Unknown Item"



- Filling the missing Price:

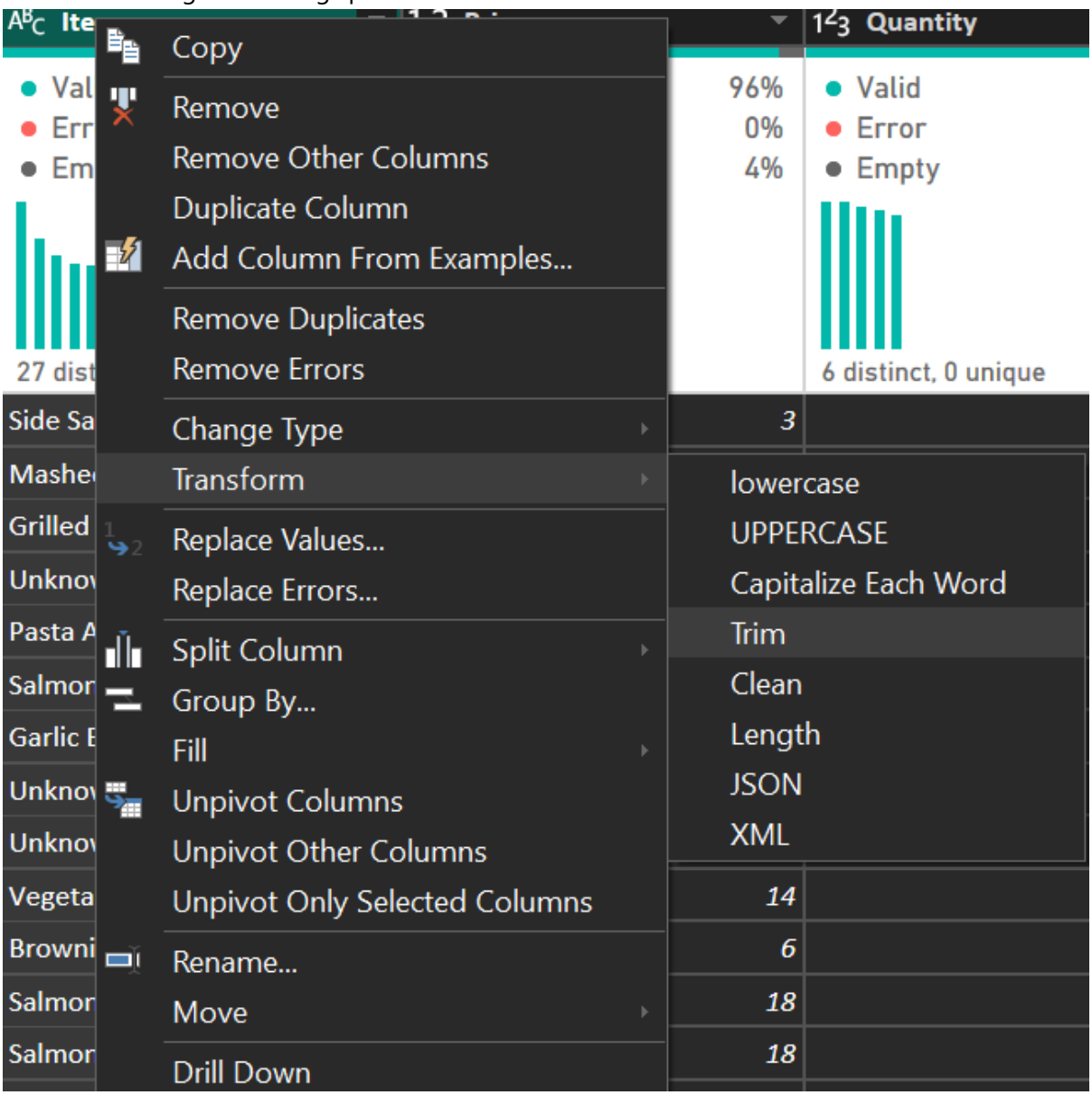


- Filled missing Quantity with 1 because it is reasonable to assume that if an item was ordered, at least one unit was purchased.

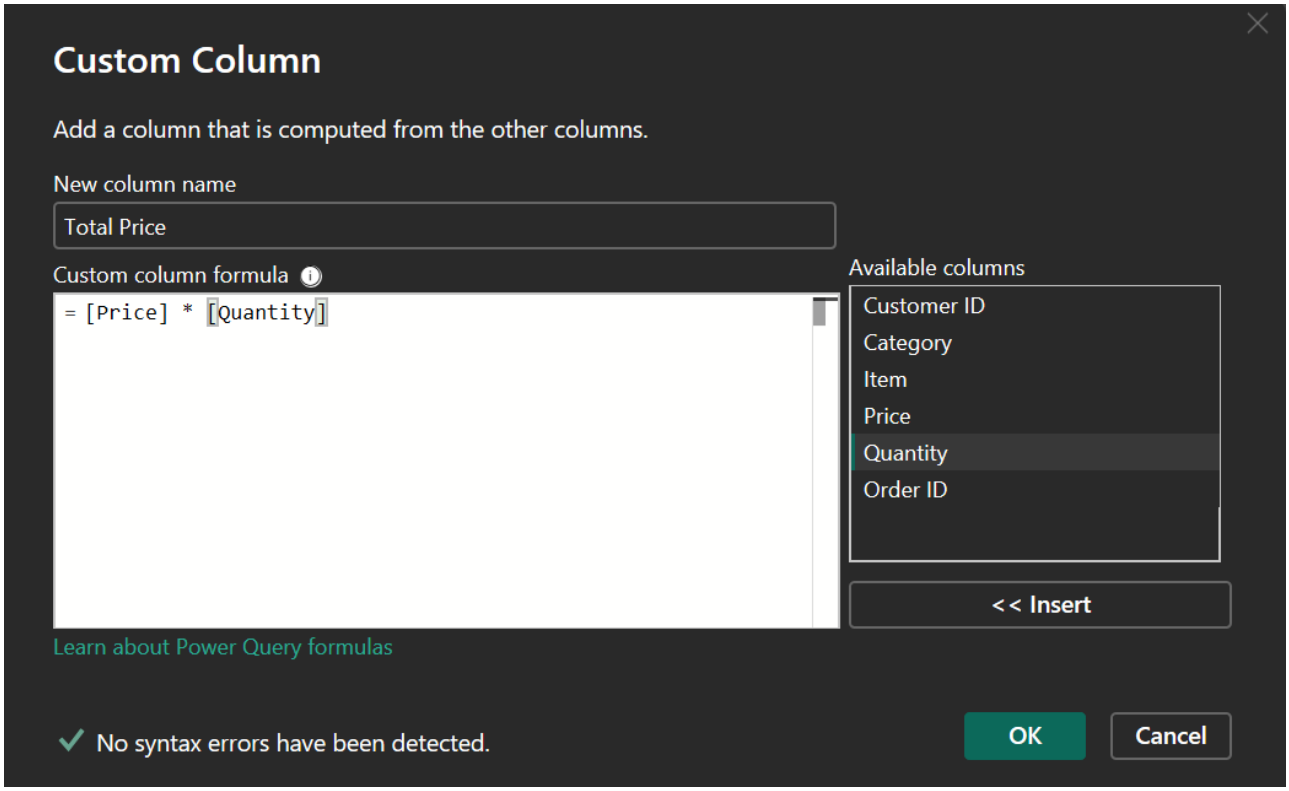


- Standardizing inconsistent data entries

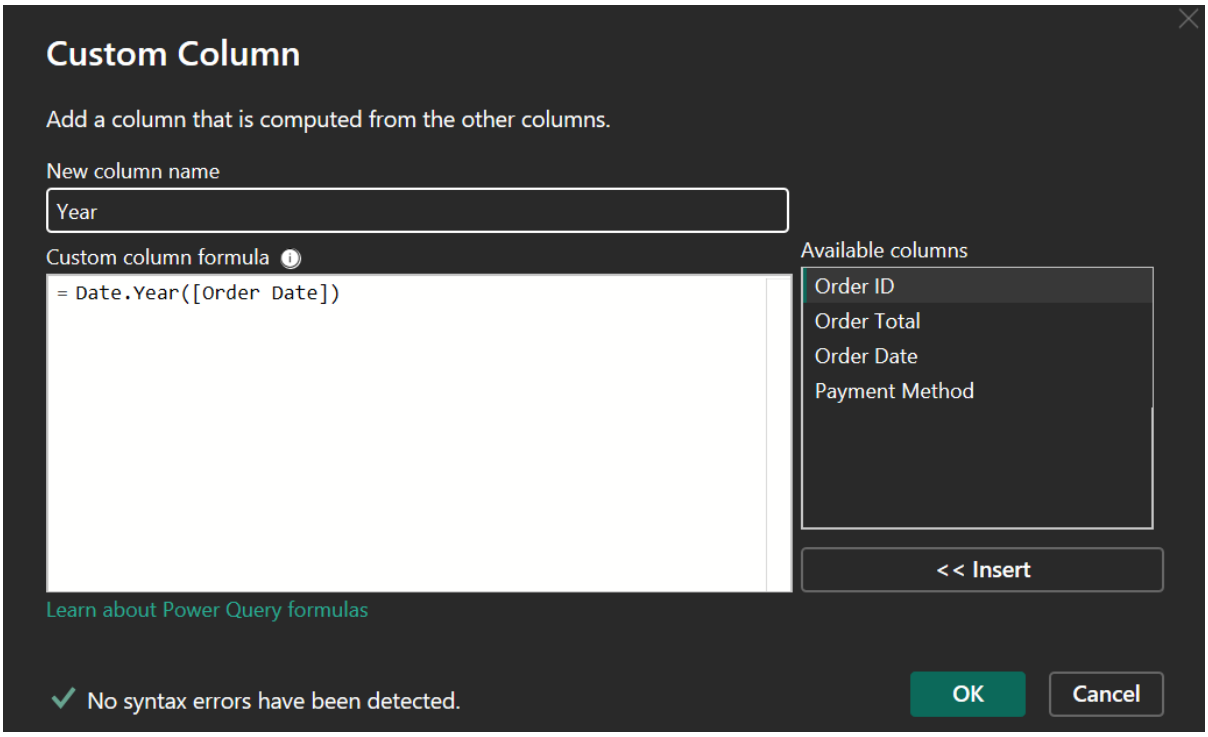
- Removed leading and trailing spaces in the Item column



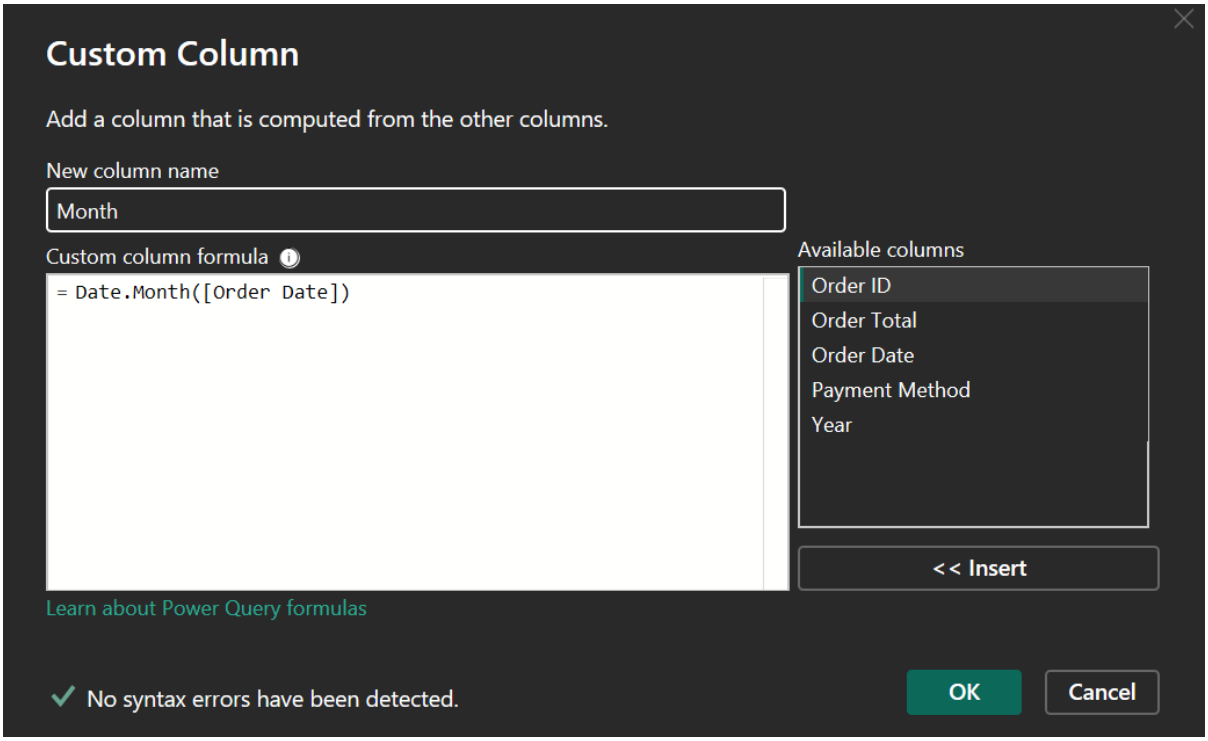
- Creating derived columns: `Total Price = (Price × Quantity)` in the Customer Table



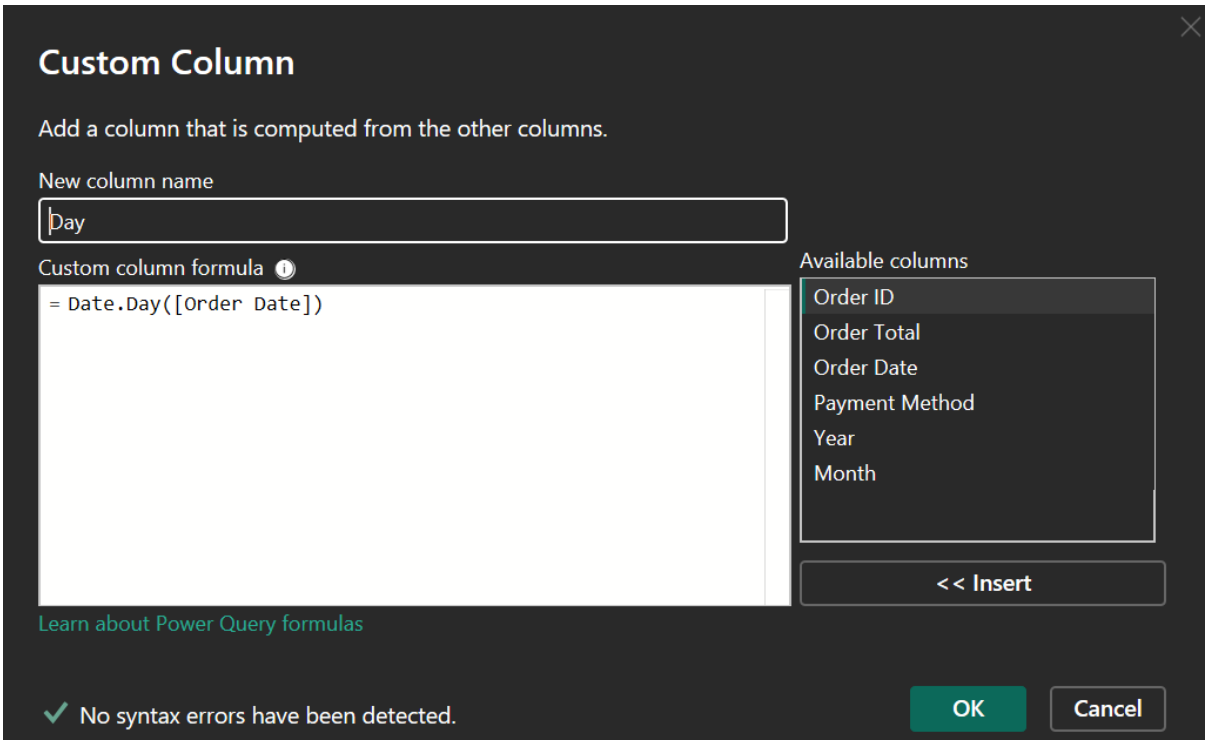
- Extracting:
  - Year from the Order Date column to analyze sales trends over time



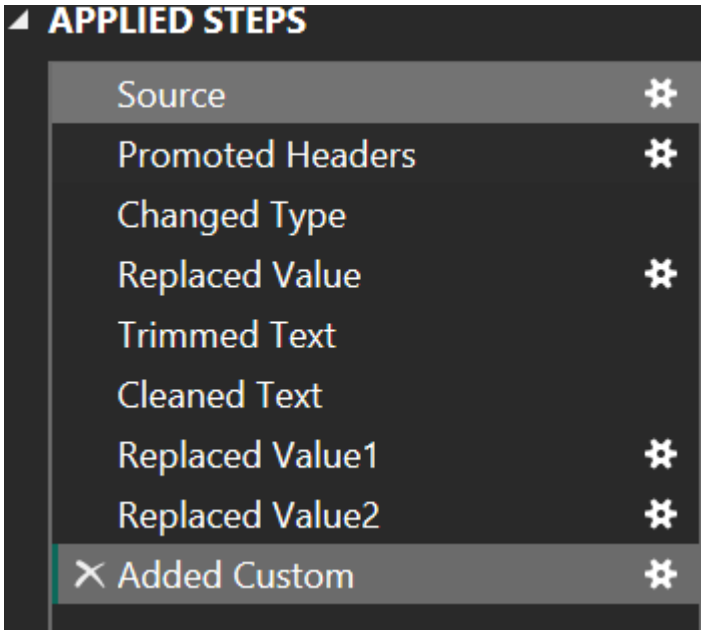
- Month from the Order Date column to identify seasonal patterns in sales

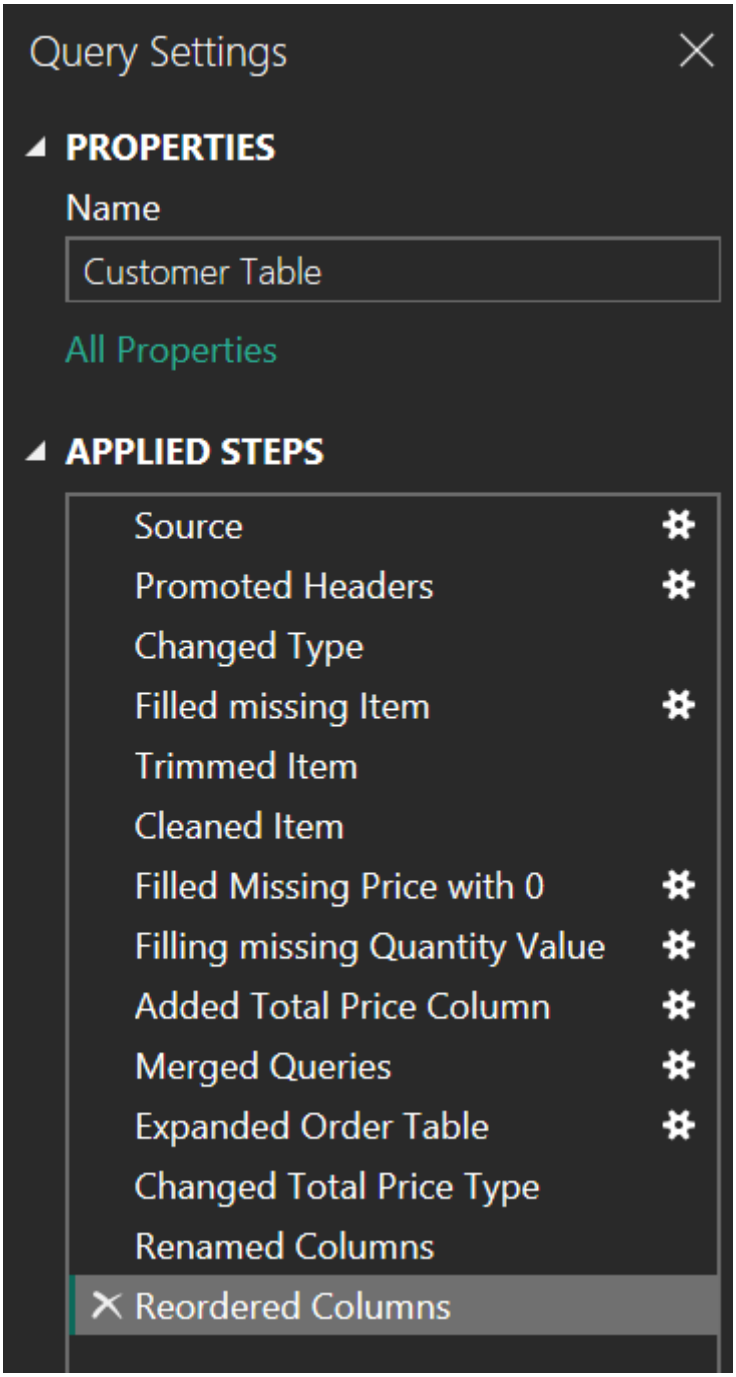


- Day of the week from the Order Date column to determine which days are busiest for the restaurant

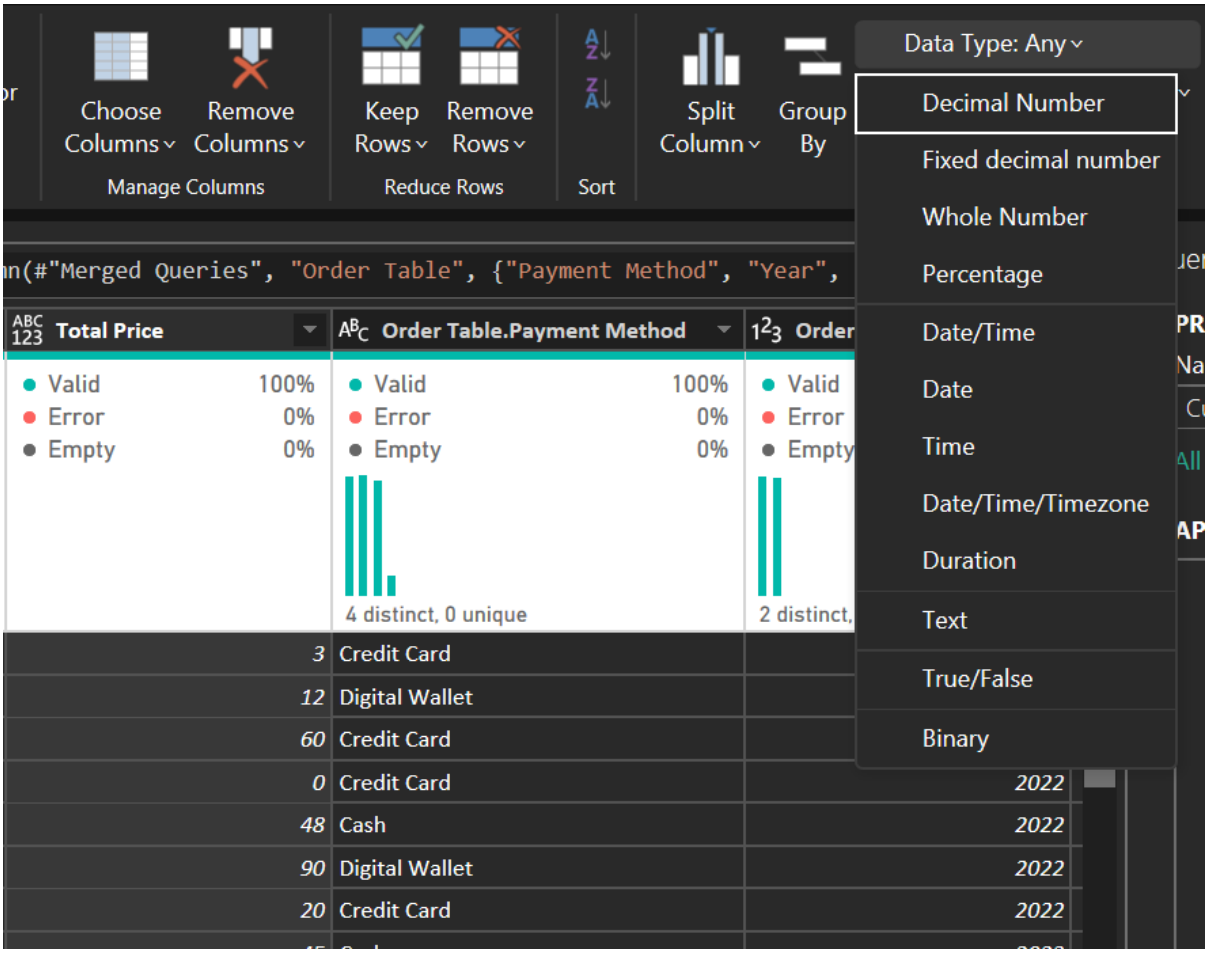


- Renaming queries professionally for better readability and organization

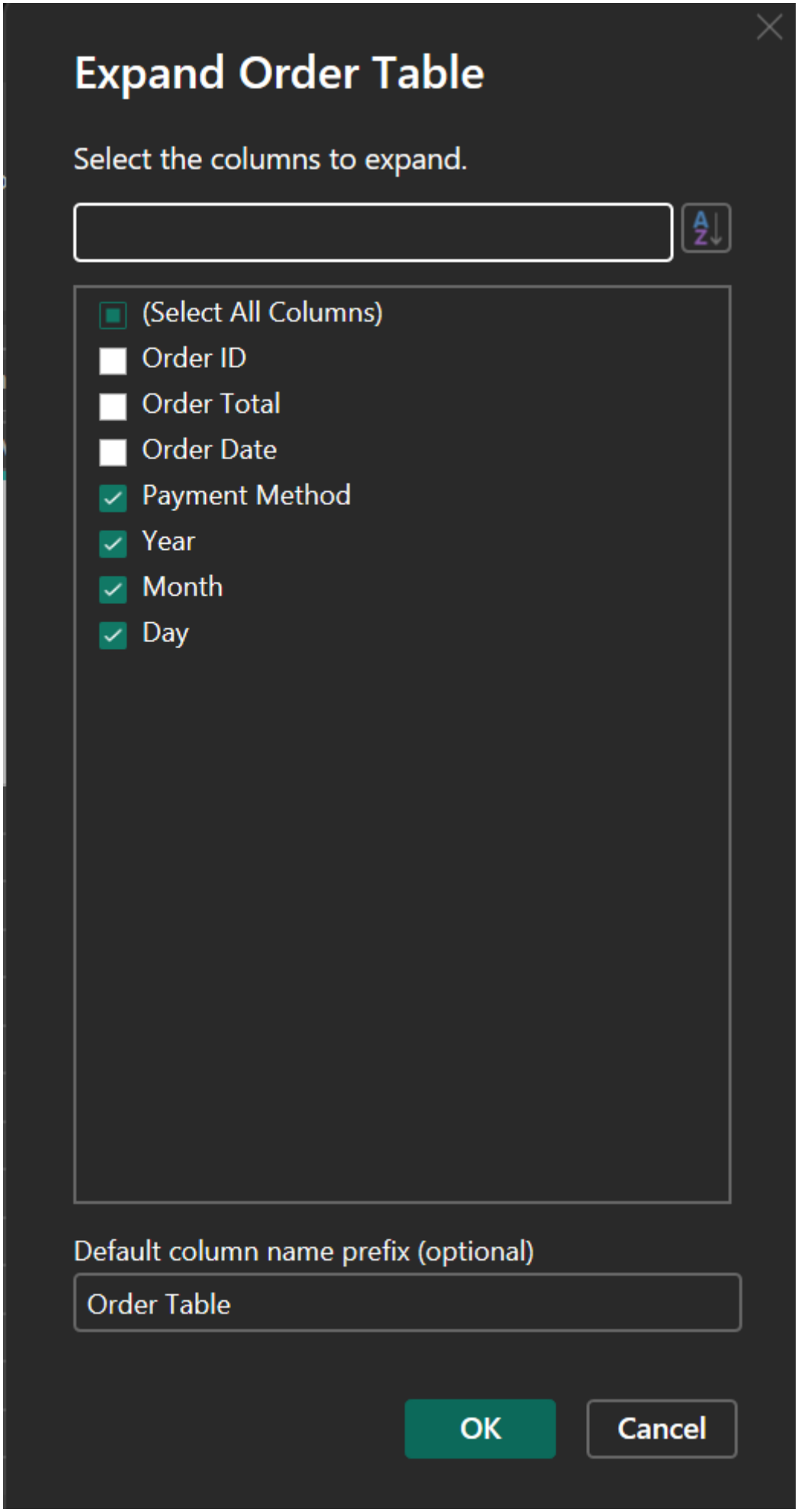
- Before Renaming 

- After Renaming 

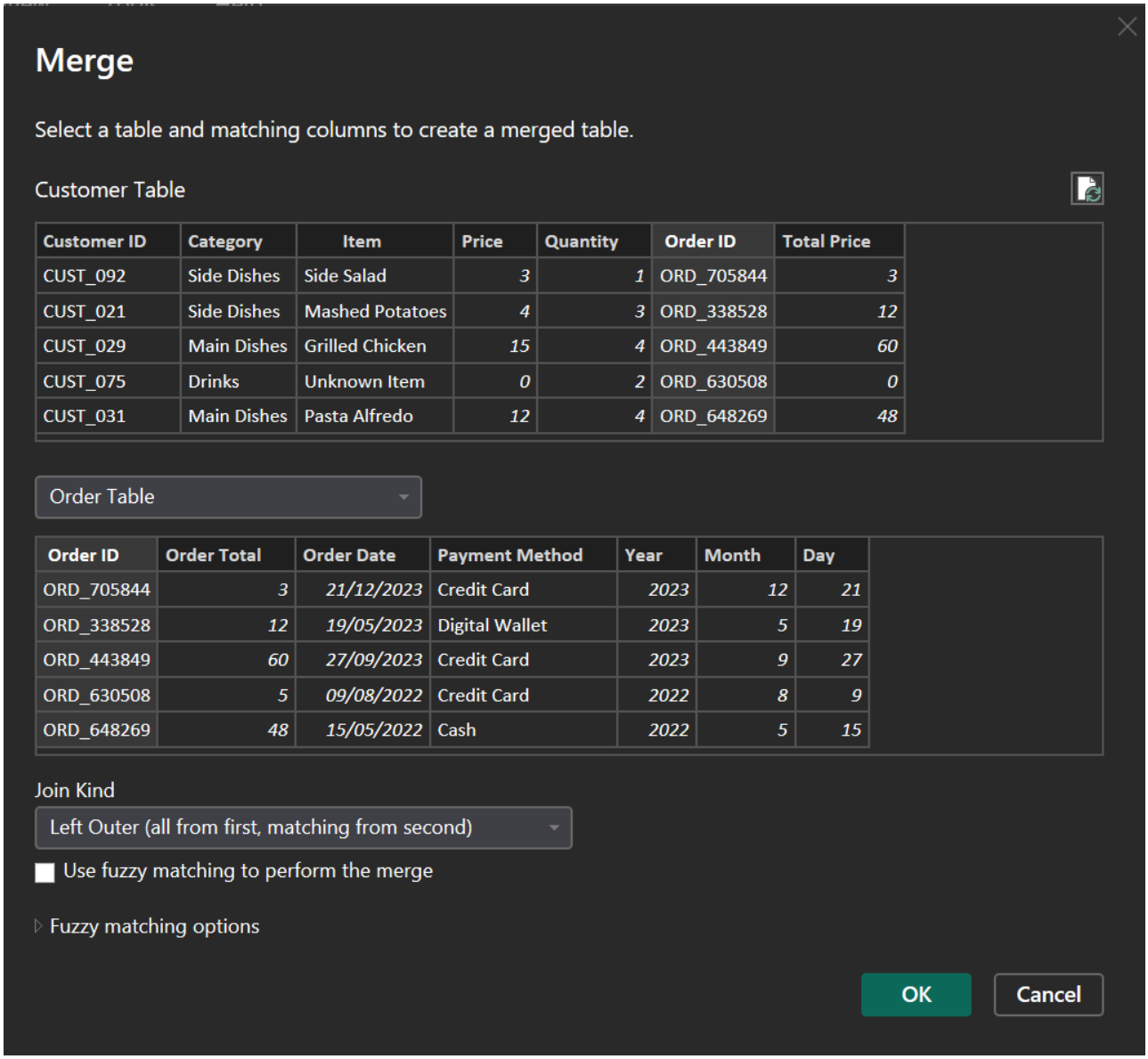
- Fixing data types:
  - Changed Total Price to Decimal Number from Any



- Removing unnecessary columns that do not contribute to the analysis



- Merging the two tables using the Order ID column to create a comprehensive dataset for analysis



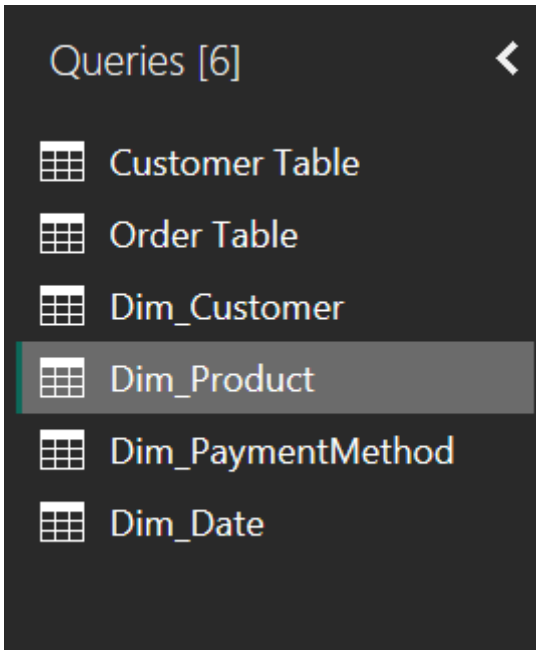
Customer ID	Category	Item	Price	Quantity	Order ID	Total Price
CUST_092	Side Dishes	Side Salad	3	1	ORD_705844	3
CUST_021	Side Dishes	Mashed Potatoes	4	3	ORD_338528	12
CUST_029	Main Dishes	Grilled Chicken	15	4	ORD_443849	60
CUST_075	Drinks	Unknown Item	0	2	ORD_630508	0
CUST_031	Main Dishes	Pasta Alfredo	12	4	ORD_648269	48

Order ID	Order Total	Order Date	Payment Method	Year	Month	Day
ORD_705844	3	21/12/2023	Credit Card	2023	12	21
ORD_338528	12	19/05/2023	Digital Wallet	2023	5	19
ORD_443849	60	27/09/2023	Credit Card	2023	9	27
ORD_630508	5	09/08/2022	Credit Card	2022	8	9
ORD_648269	48	15/05/2022	Cash	2022	5	15

Part B: Data Modelling

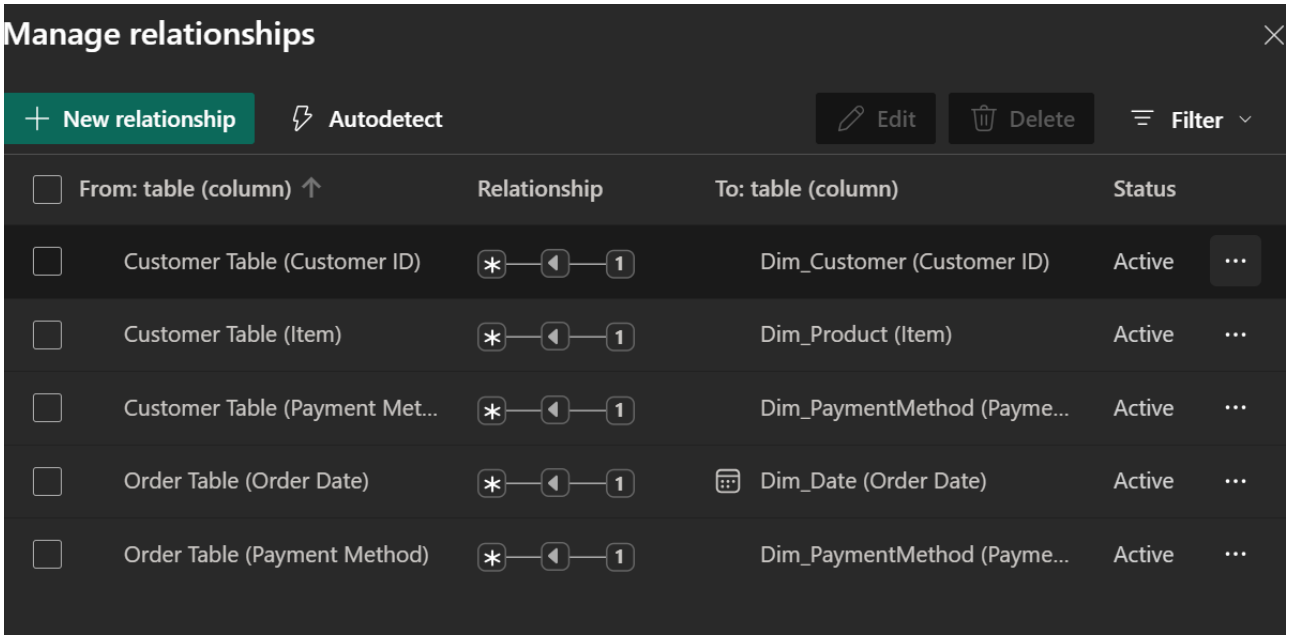
Design a clean and logical analytical data model.

- Create relationships between the tables.



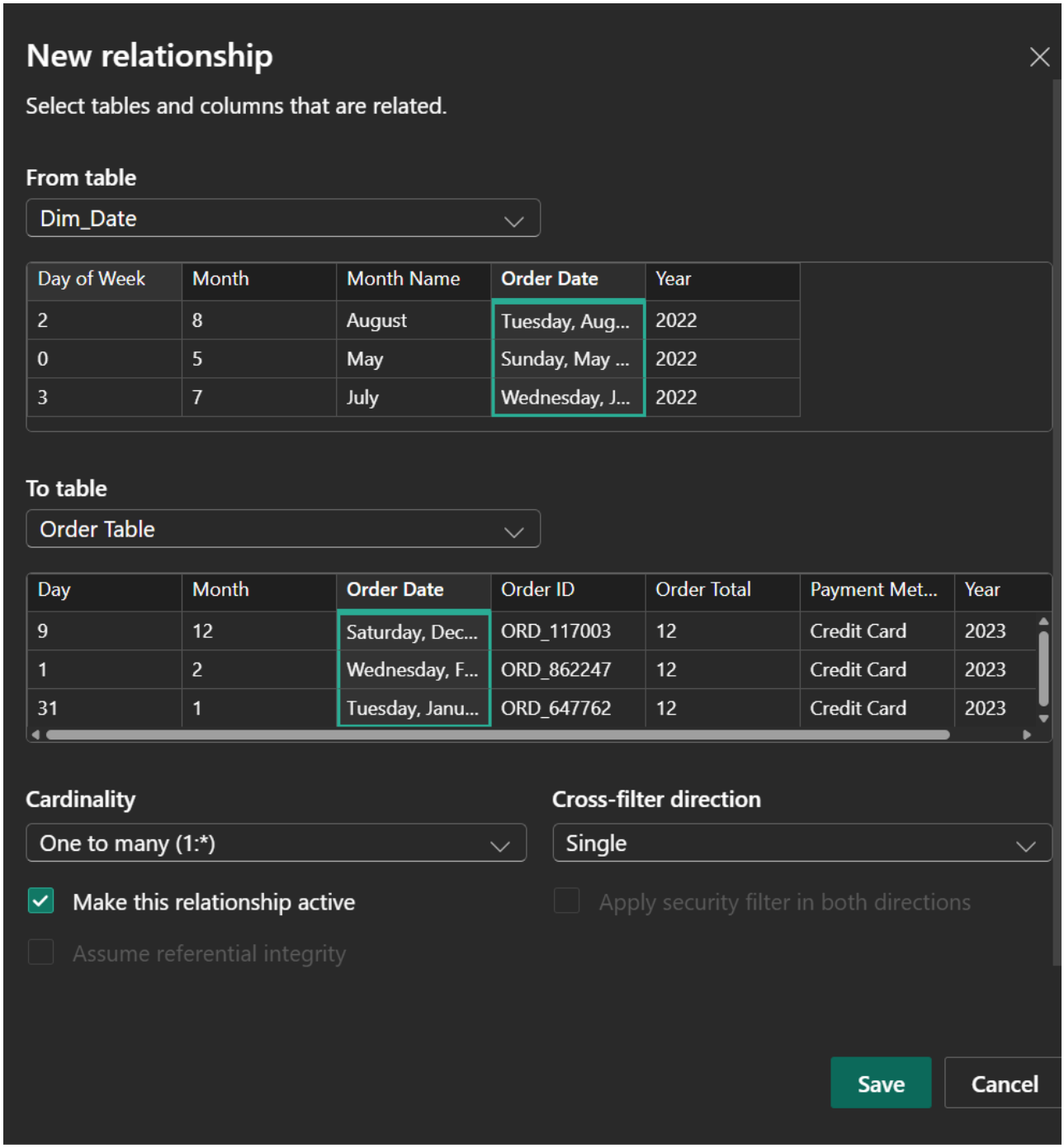
- Creating Dimention table for Order Date
  - Dim_Date
    - Date (unique)
    - Year, Month Number, Month Name, Day Name, Day of Week, etc.
  - Dim_Customer
    - Customer ID (unique)
  - Dim_Product
    - Item (unique)
    - Category
  - Dim_PaymentMethod
    - Payment Method (unique)
- Create relationships between tables
- Implement a Star Schema structure
- Apply correct cardinality
  - Dim_Product (1) → Fact_OrderItems (*)
  - Dim_Customer (1) → Fact_OrderItems (*)
  - Order Table (1) → Fact_OrderItems (*) via Order ID
  - Dim_Date (1) → Order Table (*)
  - Dim_PaymentMethod (1) → Order Table (*)

- Avoid many-to-many relationships

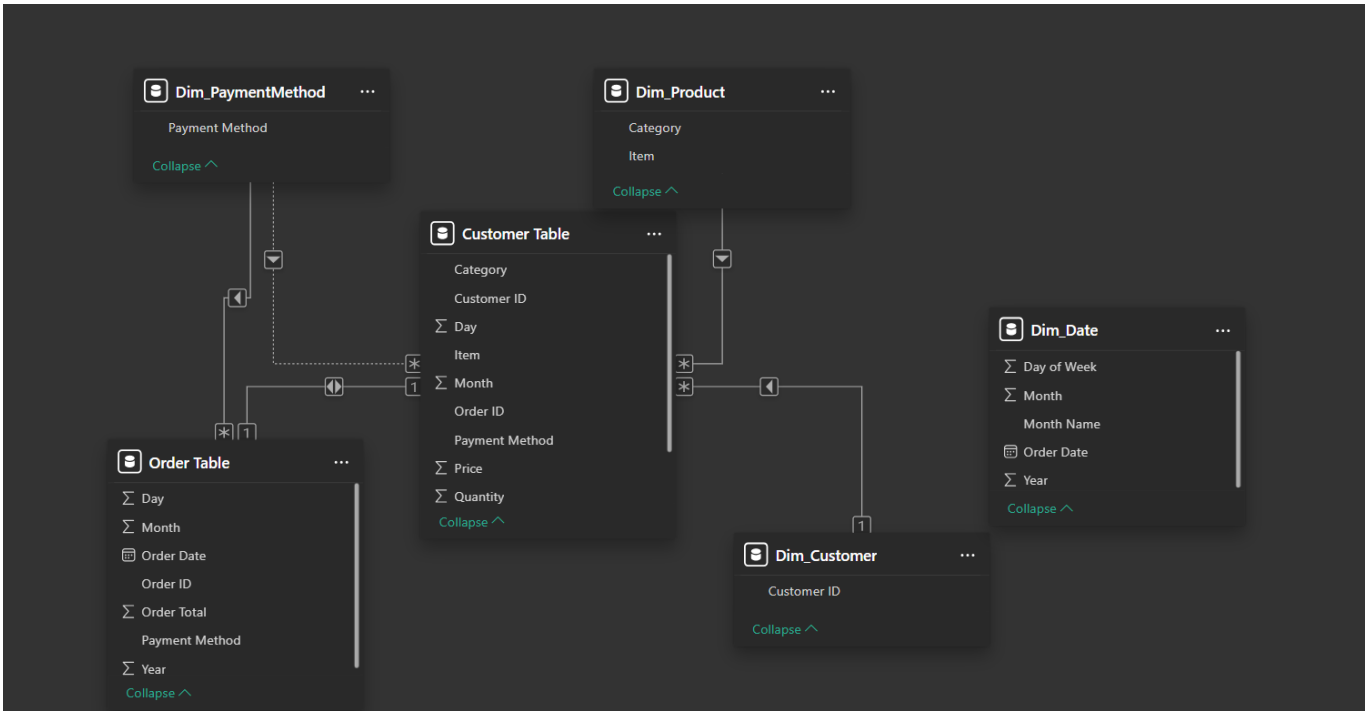


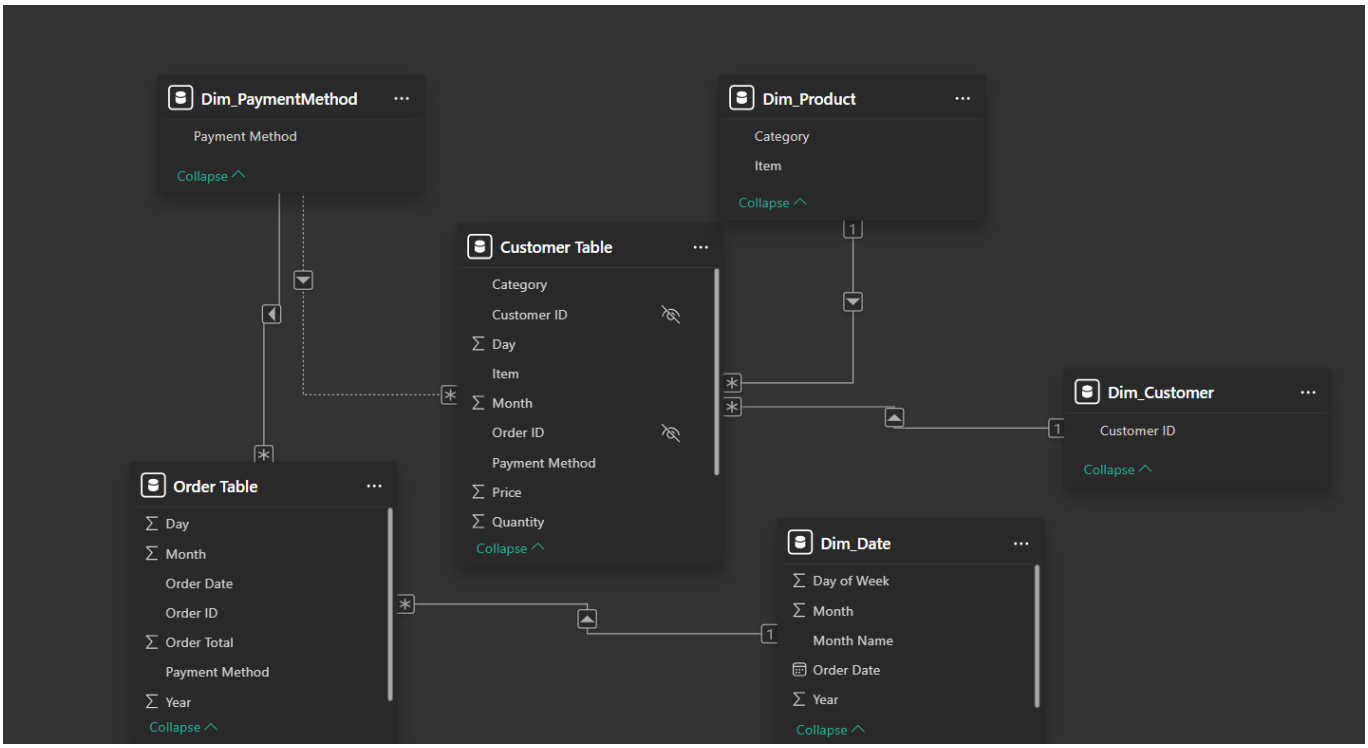
- Ensure proper filter direction
- Hide unnecessary technical fields

- Organize tables logically within Model View



Model View after creating relationships and dimension tables:



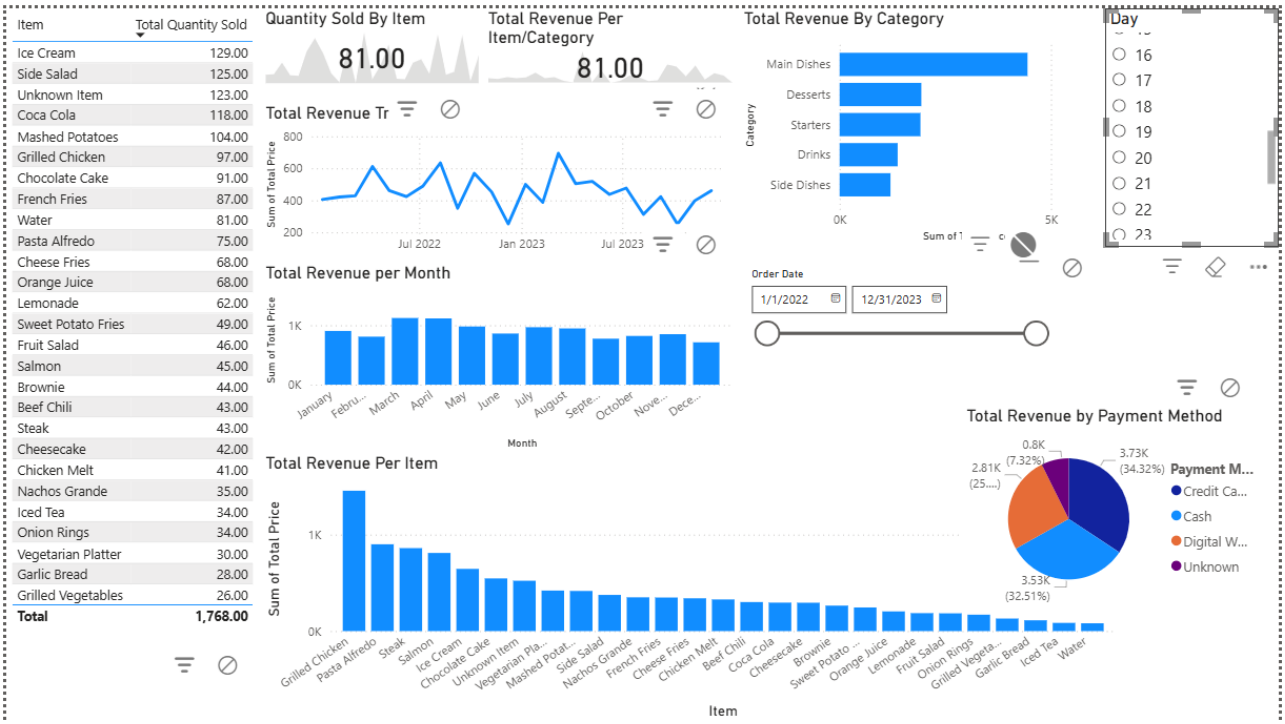


Part C: Dashboard Development

Trend Analysis:

- Created a line chart to show the trend of total sales over time (by month and year).
- Created a pie chart to show the distribution of Payment Methods used by customers.
- Created a histogram to show the distribution of Total Revenue per Item and per month.
- Create a bar chart to show total revenue by category.

- Created a table to show the top 10 best-selling items based on quantity sold.



Links to the report and dashboard in Power BI Service

To view the full report, please click to get the interactivity here: https://app.powerbi.com/links/1-WjQTQD5N?ctid=16d83ee6-254a-469d-a6cc-54e2ca2313e7&pbi_source=linkShare

Click here to view the report in Power BI Service: https://app.powerbi.com/groups/me/reports/09309554-c21b-461b-ae29-5672ae075509/3f32b93068d2b8ee210e?experience=power-bi

Click here to view the dashboard in Power BI Service: Dashboard Link: https://app.powerbi.com/groups/me/dashboards/219420dc-7d69-46b2-aeb8-06b325cf2101?ctid=16d83ee6-254a-469d-a6cc-54e2ca2313e7&pbi_source=linkShare

Note, the dashboard link will only work when I have given you access to the dashboard. Please let me know if you want to view the dashboard, and I will give you access.

Click here to view the Github repository for the Power BI project: GitHub Repository: https://github.com/Calvin-Gacheru/Business-Intelligence-And-Visualization